

Instructor Key: Lab 1 – Gridworld and the Agent–Environment Loop

AAI 4323 – Reinforcement Learning

Overview

This document provides the complete instructor solution key for Lab 1. It includes the correct behavior of the Gridworld environment, expected agent trajectories, cumulative returns, and common student errors.

The environment:

```
S . . .  
. # . .  
. . . .  
. . T G
```

Rewards:

- Step: -1
- Trap (T): -10 (terminal)
- Goal (G): $+10$ (terminal)

Transitions are deterministic.

Correct Environment Behavior

The environment must implement:

- Deterministic transitions based on the chosen action.
- Boundary checks: actions that move into walls or outside the grid leave the agent in place.
- Terminal states: stepping on T or G ends the episode.
- Rewards:
 - -1 for each non-terminal step.
 - $+10$ for reaching G.
 - -10 for reaching T.

Expected Agent Behavior

Students implement a **random agent**:

- Uniform random action selection from $\{\uparrow, \rightarrow, \downarrow, \leftarrow\}$.
- No memory or learning.
- Tracks cumulative reward.

Expected Episode Traces

Because the agent is random, traces vary. However, the following patterns are typical and should appear in correct implementations:

Example Episode 1 (Agent reaches the trap)

States:

(0,0) -> (0,1) -> (1,1) [wall, stays] -> (2,1) -> (3,1) -> (3,2) [T]

Actions:

Right, Down, Right, Down, Down

Rewards:

-1, -1, -1, -1, -10

Total Return: -14

Example Episode 2 (Agent reaches the goal)

States:

(0,0) -> (0,1) -> (0,2) -> (0,3) -> (1,3) -> (2,3) -> (3,3) [G]

Actions:

Right, Right, Right, Down, Down, Down

Rewards:

-1, -1, -1, -1, -1, +10

Total Return: +5

Example Episode 3 (Agent wanders without reaching terminal)

States:

(0,0) -> (0,0) -> (1,0) -> (1,0) -> (2,0) -> (2,1) -> (2,2)

Actions:

Up, Down, Left, Down, Right, Right

Rewards:

-1, -1, -1, -1, -1, -1

Total Return: -6

Episodes should terminate only when reaching T or G.

Expected Rendering

A correct rendering function should produce output similar to:

```

S . . .
. # . .
. A . .
. . T G

```

The agent's position updates each step.

Instructor Notes

- Students often forget to prevent movement into walls; the agent should remain in place.
- Many students incorrectly give 0 reward for terminal states; the correct rewards are +10 and -10.
- Some students forget to terminate the episode after reaching T or G.
- A few students mistakenly allow diagonal movement; only 4 actions are allowed.
- If cumulative returns are always -1, the student likely forgot terminal rewards.

Expected Output Characteristics

Correct implementations should show:

- Highly variable episode lengths.
- Frequent visits to the trap due to random wandering.
- Occasional successful paths to the goal.
- Cumulative returns typically between -20 and +5.

Rubric (Instructor Reference)

Component	Points
Environment Implementation	30
Agent Implementation	20
Episode Execution & Logging	25
Visualization	15
Code Quality & Organization	10
Total	100

Summary

A correct Lab 1 submission demonstrates:

- A functioning deterministic Gridworld.
- A random agent interacting through the agent-environment loop.
- Correct reward accumulation.
- Proper episode termination.
- Clear printed trajectories and visualizations.

These foundations are essential for Lab 2 and all subsequent reinforcement learning labs.